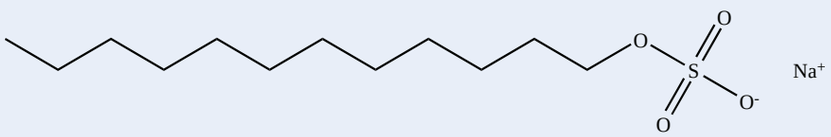
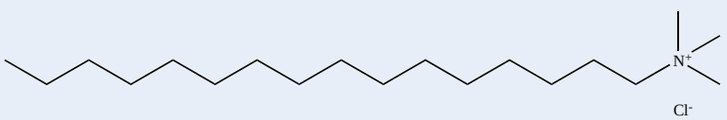
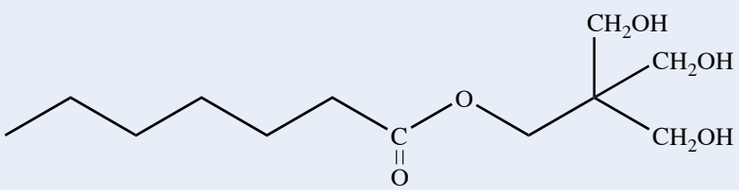
same time the negatively charged carboxylate groups are strongly attracted by water, thus leading to the formation of small droplets called micelles and grease is floated away from the solid object. When the water is rinsed away, the grease goes with it. As a result, the cloth gets free from dirt and the droplets are washed away with water. The micelles do not combine into large drops because their surfaces are all negatively charged and repel each other. The cleansing ability of a soap depends upon its tendency to act as a emulsifying agent between water and water insoluble greases.

15.3.2 Detergents:

Synthetic detergents are formulated products containing either sodium salts of alkyl hydrogen sulphates or sodium salts of long chain alkyl benzene sulphonic acids. There are three types of detergents.

Detergent Type	Example
Anionic detergent	Sodium Lauryl sulphate (SLS) 
Cationic detergent	n-hexadecyltrimethyl ammonium chloride  *N,N,N*-trimethylhexadecan-1-aminium chloride
Non-ionic detergent	Pentaerythrityl stearate.  3-hydroxy-2,2-bis(hydroxymethyl)propyl heptanoate

Detergents are superior to soaps as they can be used even in hard water and in acidic conditions. The cleansing action of detergents are similar to the cleansing action of soaps.

15.4 Polymers

The term Polymer is derived from the Greek word 'polumeres' meaning "having many parts". The constitution of a polymer is described in terms of its structural units called monomers. Polymers consists of large number of monomer units derived from simple molecules. For example: PVC(Poly Vinyl Chloride). is a polymer which is obtained from the monomer vinyl chloride. Polymers can be classified based on the source of availability, structure, molecular forces and the mode of synthesis. The following chart explain different classification of polymers.

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